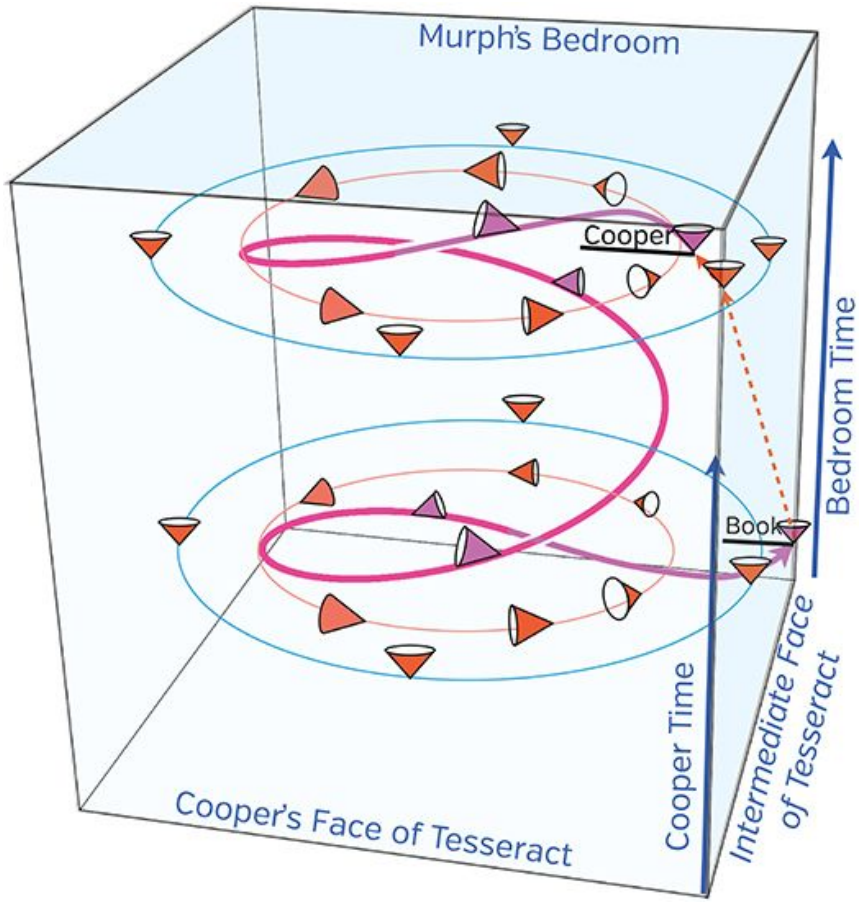


Figure TN.1. The causal structure of spacetime inside the tesseract with one space dimension omitted.

Can you understand from this diagram how it is that the gravitational-wave message travels at the speed of light, yet moves backward relative to bedroom time and Cooper's time? And can you understand how, by contrast, the light ray travels at the speed of light and moves forward relative to bedroom time and Cooper's time? Compare with our discussion of Escher's drawing, Figure 30.6.

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60 This Paczynski-Wiita modification of gravity was used in developing the black hole's influence on spacecraft orbits for a gravitational-slingshot video game associated with